

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****CERTIFICATE OF INSTALLATION****Note:** This table completed by **HERSECC** Registry.

Project Name:	Enforcement Agency:
Dwelling Address:	Permit Number:
City and Zip Code:	Permit Application Date:

A. Pre-Cooling Verification

01	02	03
SC System ID/Name from CF1R	Thermostat Type	Pre-Cooling

B. Mandatory Requirements for Occupant Controlled Smart Thermostats

Below is a list of required functional specifications and behaviors. Consult Joint Appendix JA5 for more information.

01	Setback Capabilities. An OCST shall meet the requirements of Section 110.2(c). Thermostats for heat pumps shall also meet the requirements of Section 110.2(b).
02	Restart Settings. In the event of a disruption of power to the device that results in power off or restart, upon device restart, the device shall automatically restore the most recently programmed settings, including reconnection to a network, if the device was previously enabled and network connectivity is available.
03	Automatic Rejoin. AN OCST shall connect, and remain connected in its communication path and control end point. The OCST shall incorporate an automatic rejoin function. When physical and/or logical communication is lost, the OCST shall trigger its automatic rejoin function to restore the physical and/or logical communication.
04	Event Responses. Event response, unless overridden by the occupant or modified by an energy management control system or service, may be triggered by price signals or Demand Response Signals. The OCST shall provide one set of event responses for price signals and one set of event responses for Demand Response Signals. The responses may be common for both types of events.
05	User Display and Interface. The OCST shall have the capability to display information to the user.
06	Require Functional Behavior. (a) <i>Normal Operation.</i> Normal operation of an OCST is defined to be the OCST's prevailing mode of operation as determined by the occupant's prior settings and use of features provided by the OCST manufacturer's design. Aspects of normal operation of an OCST may be modified or interrupted in response to occupant subscribed price signals of when Demand Response Periods are in progress, but only to the extent specified by the occupants or their representatives. (b) <i>Demand Responsive Control.</i> Upon receiving a price signal or a Demand Response Signal, OCSTs shall be capable of automatic event response by adjusting the currently applicable temperature set point by the number of degrees indicated in the temperature offset (heating or cooling, as appropriate).
07	HVAC System Interface. HVAC wiring terminal designations shall be clearly labeled. OCSTs shall use labels that comply with Table 5-1 in NEMA DC 3-2013.

C. Compliance Statement

--

Registration Number:

Registration Date/Time:

HERSECC

Provider:

CA Building Energy Efficiency Standards - ~~2022~~2025 **Residential-Single Family** ComplianceJanuary ~~2022~~2025

**SAMPLE FORM – NOT VALID FOR SUBMISSION TO BUILDING DEPARTMENTS****DOCUMENTATION AUTHOR'S DECLARATION STATEMENT**

1. I certify that this Certificate of Installation documentation is accurate and complete.

Documentation Author Name:	Documentation Author Signature:
Documentation Author Company Name:	Date Signed:
Address:	CEA/ <u>HERSECC</u> Certification Identification (If applicable):
City/State/Zip:	Phone:

RESPONSIBLE PERSON'S DECLARATION STATEMENT

I certify the following under penalty of perjury, under the laws of the State of California:

1. The information provided on this Certificate of Installation is true and correct.
2. I am either: a) a responsible person eligible under Division 3 of the Business and Professions Code in the applicable classification to accept responsibility for the system design, construction, or installation of features, materials, components, or manufactured devices for the scope of work identified on this Certificate of Installation, and attest to the declarations in this statement, or b) I am an authorized representative of the responsible person and attest to the declarations in this statement on the responsible person's behalf.
3. The constructed or installed features, materials, components or manufactured devices (the installation) identified on this Certificate of Installation conforms to all applicable codes and regulations and the installation conforms to the requirements given on the Certificate of Compliance, plans, and specifications approved by the enforcement agency.
4. I understand that a registered copy of this Certificate of Installation shall be posted or made available with the building permit(s) issued for the building, and made available to the enforcement agency for all applicable inspections, and I will take the necessary steps to ensure this requirement is accomplished.
5. I understand that a registered copy of this Certificate of Installation is required to be included with the documentation the builder provides to the building owner at occupancy, and I will take the necessary steps to ensure this requirement is accomplished.

Responsible Builder/Installer Name:	Responsible Builder/Installer Signature:	
Company Name: (Installing Subcontractor or General Contractor or Builder/Owner)	Position With Company (Title):	
Address:	CSLB License:	
City/State/Zip:	Phone:	Date Signed:

Registration Number:

Registration Date/Time:

HERSECC

Provider:

CA Building Energy Efficiency Standards - 20222025 Residential-Single Family ComplianceJanuary 20222025

CF2R-MCH-34-E User Instructions

A. Pre-Cooling Verification

1. This field is filled out automatically. It is referenced from the Certificate of Compliance (CF1R), which must be completed prior to this document.
2. This field is filled out automatically. It is referenced from the Certificate of Installation (CF2R-MCH-01), which must be completed prior to this document.
3. States “capable” for all systems listed.

B. Mandatory Requirements for Occupant Controlled Smart Thermostats

This table lists the requirements for all occupant controlled smart thermostats. Installer must ensure all the requirements in this table are met.

C. Compliance Statement

To comply, the total installed whole house fan airflow must equal to or greater than the required airflow and the installed fan efficacy must be less than or equal to the required fan efficacy.

A Pre-Cooling Verification

<<require one row for each SC System identified on the CF1R>>

01	02	03
SC System ID/Name from CF1R	Thermostat Type	Pre-Cooling
<<Auto populate from CF1R>>	<<Auto populate from CF2R-MCH-01a field D08 SC System Thermostat Type >>	<<If CF1R-PRF indicates required special feature = Pre-Cooling, AND field A02 = Occupant Controlled Smart Thermostat (OCST), then report "Complies"; Else report "Does Not Comply" >

B. Mandatory Requirements for Occupant Controlled Smart Thermostats

Below is a list of required functional specifications and behaviors. Consult Joint Appendix JA 5 for more information.

01	Setback Capabilities. An OCST shall meet the requirements of Section 110.2(c). Thermostats for heat pumps shall also meet the requirements of Section 110.2(b).
02	Restart Settings. In the event of a disruption of power to the device that results in power off or restart, upon device restart, the device shall automatically restore the most recently programmed settings, including reconnection to a network, if the device was previously enabled and network connectivity is available.
03	Automatic Rejoin. AN OCST shall connect, and remain connected in its communication path and control end point. The OCST shall incorporate an automatic rejoin function. When physical and/or logical communication is lost, the OCST shall trigger its automatic rejoin function to restore the physical and/or logical communication.
04	Event Responses. Event response, unless overridden by the occupant or modified by an energy management control system or service, may be triggered by price signals or Demand Response Signals. The OCST shall provide one set of event responses for price signals and one set of event responses for Demand Response Signals. The responses may be common for both types of events.
05	User Display and Interface. The OCST shall have the capability to display information to the user.
06	Require Functional Behavior. (c) <i>Normal Operation.</i> Normal operation of an OCST is defined to be the OCST's prevailing mode of operation as determined by the occupant's prior settings and use of features provided by the OCST manufacturer's design. Aspects of normal operation of an OCST may be modified or interrupted in response to occupant subscribed price signals of when Demand Response Periods are in progress, but only to the extent specified by the occupants or their representatives. (d) <i>Demand Responsive Control.</i> Upon receiving a price signal or a Demand Response Signal, OCSTs shall be capable of automatic event response by adjusting the currently applicable temperature set point by the number of degrees indicated in the temperature offset (heating or cooling, as appropriate).
07	HVAC System Interface. HVAC wiring terminal designations shall be clearly labeled. OCSTs shall use labels that comply with Table 5-1 in NEMA DC 3-2013.

C. Compliance Statement

<<If all SC System in Section A. qualify for Pre-Cooling (A03 = Complies), then report **"Building Complies – "All SC systems are controlled by occupant controlled smart thermostats which meet the requirements of JA5 in the Reference Appendices"**

Else report **"Building Fails – Pre-cooling verification requirements have not been met">>**